Roll No. .....

# TBT–405(a)

## B. TECH. (BIOTECHNOLOGY WITH SPECIALIZATION IN COMPUTER SC. & BIOLOGY) (FOURTH SEMESTER) MID SEMESTER EXAMINATION, March, 2024

WASTE MANAGEMENT AND UPCYCLING

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the following with the help of data : Global Waste generation, its type and annual regional waste generation. (CO1)

OR

(b) What are the drivers of waste generation and what is the need for waste upcycling ? (CO1)

2. (a) Give schematic diagram of waste treatment. (CO1)

OR

(b) Discuss the operation and function of Effluent Treatment Plant. (CO1)

3. (a) Explain the structure and functions of waste water stabilization ponds. (CO2)

OR

(b) Write short notes on the following : (CO2)

(i) Aerosol and Particulate Matter

(ii) Reed beds in waste water management

*P. T. O.*

4. (a) Give a detailed account of air pollution and its impacts and discuss any approach to tackle air pollution. (CO2)

OR

(b) Write short notes on the following : (CO2)

(i) Electrostatic Precipitators

(ii) Carbon Sequestration

5. (a) Discuss the thermal technologies for the treatment of waste. (CO3)

OR

(b) What is the purpose of incineration in waste treatment ? Give example in support of your answer. (CO3)



50

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# TBT–405(C)

## B. TECH. (BIOTECHNOLOGY WITH SPECIALIZATION IN COMPUTER SC. & BIOLOGY) (FOURTH SEMESTER) MID SEMESTER EXAMINATION, March, 2024

BIOMATERIALS

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Write and explain the types of generation of biomaterial. (CO1, CO1)

OR

(b) What are major differences between polymer, ceramics, and metal ? Write also the chemical bond of the respective. (CO1, CO1)

2. (a) White some applications of polymer composite in the field of biomaterials with their utility. (CO1, CO1)

OR

(b) Define the following on the basic of constituents, mechanical and tribological properties. (CO1, CO1)

(i) Plant fiber

*P. T. O.*

(ii) Animal fiber

3. (a) Write a short notes on following : (CO1, CO1)

(i) bulk modulus,

(ii) malleability

(iii) toughness

(iv) refractoriness

(v) modulus of rigidity.

Write the formula also if required.

OR

(b) Explain with the help of examples, the bio ceramics along with its medical applications. (CO1, CO1)

4. (a) What is nucleation and mechanism of growth in a structure ? How the rate of cooling effects the grain and gain boundary formation ? Explain in Brief. (CO1, CO2)

OR

(b) What would be the greatest length of a steel wire which when fixed at one end can hand freely without breaking ? Density of steel = 6500 kg/m$^3$. Breaking stress for steel $= 8.1 \times 10^8 \text{N/m}^2$. (CO1, CO2)

5. (a) A wire 4 m long and 4 mm in diameter, when stretched by weight of 12 kg has its length increased by 0.4 mm. Find the stress, strain and Young's modulus of the material of the wire. $g = 9.8 \text{ m/s}^2$. (CO2, CO2)

OR

(b) What must be the elongation of a wire 8 m long so that the strain is 2% ? If the wire has cross-selection of 2 mm$^2$ and is stretched by 20 kg-wt, what is the stress ? (CO2, CO2)

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# TBT–401

## B. TECH. (BIOTECHNOLOGY WITH SPECIALIZATION IN COMPUTER SC. & BIOLOGY) (FOURTH SEMESTER) MID SEMESTER EXAMINATION, March, 2024

BIOPHYSICS

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the electromagnetic radiation for biological function and interaction of EM radiation with matter. (CO1)

OR

(b) Discuss the ionizing radiation exposure and its biological effects with suitable examples. (CO1)

2. (a) Discuss the basic principle of spectroscopy and write the applications for biological functions. (CO1)

OR

(b) Discuss the Atomic and Molecular Absorption with suitable applications. (CO1)

***P. T. O.***

3. (a) Discuss the principle of Spectrometry and its types and biological applications. (CO2)

OR

(b) Explain the basic components of spectroscopy and its different types. (CO2)

4. (a) Discuss UV-Vis spectroscopy and electronic, vibration and rotation energy states. (CO2)

OR

(b) Explain the key parameters of Spectroscopy and with suitable examples. (CO2)

5. (a) Discuss Spectrofluorophotometer design, working, and application. (CO2)

OR

(b) Discuss the basic processes to analyze a Fluorescence Spectrum. (CO2)

Roll No. .....

# TBT–402

## B. TECH. (BIOTECHNOLOGY WITH SPECIALIZATION IN COMPUTER SC. & BIOLOGY) (FOURTH SEMESTER) MID SEMESTER EXAMINATION, March, 2024

### IMMUNOLOGY AND IMMUNOTECHNOLOGY

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are antigenic determinants ? Explain their properties and function in detail. (CO3)

OR

(b) What is Phagocytosis ? Explain the oxygen dependent and independent killing mechanism of macrophages in Phagocytosis. (CO2)

2. (a) What is acquired immune response ? Comment on its attributes and different types. (CO2)

OR

(b) What is Inflammation ? Explain its different stages of removal of pathogen as an innate immune response. (CO2)

*P. T. O.*

3. (a) Using suitable diagram, explain the structure and function of secondary lymphoid organs. (CO2)

OR

(b) Explain the significance of anatomical and physiological barriers of innate immunity. (CO2)

4. (a) What is ADCC ? Explain its significant role in elimination of damaged cells. (CO2)

OR

(b) Explain the process of antigen presentation and its significance in cell mediated immune response. (CO2)

5. (a) What are recombinant vaccines ? Explain the role of vaccine in development of acquired immunity. (CO1)

OR

(b) With suitable examples of each, differentiate between live attenuated and killed vaccine. (CO1)

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# TBT–403

## B. TECH. (BIOTECHNOLOGY WITH SPECIALIZATION IN COMPUTER SC. & BIOLOGY) (FOURTH SEMESTER) MID SEMESTER EXAMINATION, March, 2024

BIOANALYTICAL TECHNIQUES

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you understand by term bioanalytical techniques ? Discuss at least *ten* industrial applications of bioanalytical techniques. (CO1)

OR

(b) Discuss instrumentation working and application of AFM. (CO1)

2. (a) Differentiate between fluorescence microscopy and confocal microscopy. (CO1)

OR

(b) Differentiate between SEM and TEM. (CO1)

3. (a) Draw well labeled structure of compound microscope. Explain the working principle of compound microscope. (CO1, CO2)

OR

(b) Explain various factors that affect centrifugations process with the help of an equation. (CO1, CO2)

*P. T. O.*

4. (a) Write short note on the following : (CO2)

(i) Differential centrifugation, its working principle and application.

(ii) Ultracentrifugation, its working principle and application.

OR

(b) Write short note on the following : (CO2)

(i) Role of Sucrose and Glycerol in 2D SDS PAGE.

(ii) Role of ampholyte molecule in 2D SDS PAGE.

(iii) Role of SDS and PAGE in electrophoresis.

5. (a) Draw relationship between different forces acting during centrifugation process. (CO2)

OR

(b) Illustrate agarose gel electrophoresis, its working principle and application. (CO2)

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# TBT–404

## B. TECH. (BIOTECHNOLOGY WITH SPECIALIZATION IN COMPUTER SC. & BIOLOGY) (FOURTH SEMESTER) MID SEMESTER EXAMINATION, March, 2024

ENVIRONMENTAL BIOTECHNOLOGY

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the relevance of the biotechnological approaches that one should apply for the protection of the environment. (CO1, CO2)

OR

(b) Describe the methods adopted for the measurement of pollution. (CO1, CO2)

2. (a) "A form of precipitation which has high level of hydrogen ions." Explain such form and its impact with the help of diagram. (CO1)

OR

(b) Highlight the current status and the problem solving approaches for the ozone layer depletion. (CO1)

***P. T. O.***

3. (a) How will you define sludge ? Explain in detail the treatment methods involved for it. (CO2)

OR

(b) Describe the role and mechanism of thermal hydrolysis plant will the help of a diagram. (CO2)

4. (a) "A reactor which is a methane producing digester and which produces a granular sludge." Diagrammatically explain such type of reactor. (CO1, CO2)

OR

(b) "Process uses microorganisms to feed on organic contaminants in wastewater, producing a purified effluent." Explain such process diagrammatically. (CO1, CO2)

5. (a) Differentiate between trickling filter and rotating biological contractors. (CO2)

OR

(b) Describe the management strategies that should be adopted for solid waste and the processing of organic residues. (CO2)